

Roll No. | Total No. of Pages : 6

EC-2311

B. Tech. (Third Semester)

EXAMINATION, 2025-26

MATHEMATICS-III

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

1. Show that the function e^z is analytic.
2. Show that the function $f(z) = |z|^2$ is not analytic at any point.

P. T. O.

3. Evaluate :

$$\oint_C \frac{e^{2z}}{z-2} dz$$

where C is a circle $|z|=1$.

4. The probability for a rainy day in the first week September in a region in any year is assigned a value of 0.35. Calculate the probability that there will be exactly 3 rainy days.
5. If X is a Poisson variate such that $P[X=0]=0.2$, then find its variance.
6. Show that :

$$\Delta \cdot \nabla \equiv \Delta \nabla$$

7. A second degree polynomial passes through (0, 1), (1, 3), (2, 7), (3, 13). Find the polynomial.
8. Using Newton-Raphson method find a real root of $e^x \cos x = 1.2$ correct to three decimal places.

9. Evaluate :

$$\Delta^2 \left[\frac{5x+12}{x^2+5x+6} \right]$$

10. Evaluate $\int_0^1 e^{-x^2} dx$, taking $h = \frac{1}{6}$ by

Trapezoidal rule.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 12 marks. $2 \times 12 = 24$

1. Estimate the production of cotton in the year 1935 from the data given below.

Year	Production (in millions of bales)
1931	17.1
1932	13.0
1933	14.0
1934	9.6
1935	—
1936	12.4
1937	18.2

P. T. O.

2. Use Range-Kutta method of order four to find y when $x=0.4$ in steps of 0.2 given that

$$\frac{dy}{dx} = 1 + y^2$$

$$y(0) = 0.$$

3. Determine the order of convergence of Newton-Raphson method.

4. Find the residue of $\frac{z^3}{z^2 - 1}$ at $z = \infty$.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. By the method of contour integration prove that :

$$\int_0^{\infty} \frac{dx}{x^4 + a^4} = \frac{\pi\sqrt{2}}{4a^3}$$

where $a > 0$.

2. Fit a Poisson distribution to the following data and calculate the expected frequencies :

x	f
0	142
1	156
2	69
3	27
4	5
5	1

3. Find the first and second derivatives of the function tabulated below at the point $x = 1.1$.

x	$f(x)$
1	0
1.2	0.128
1.4	0.544
1.6	1.296
1.8	2.432
2.0	4.0

P. T. O.

4. Using Regula-Falsi method, compute a real root of the following equations correct to four decimal places :

(i) $e^x \sin x = 1$

(ii) $x = 3e^{-x}$

x x x x x

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MA-301/2311

B. Tech. (EC) (Third Semester)

EXAMINATION, 2023-24

MATHEMATICS—III

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any five questions. Each question carries 8 marks. $5 \times 8 = 40$

1. Show that the real and imaginary parts of the function $w = \ln z$ satisfy the Cauchy-Riemann equations when $z \neq 0$.

P. T. O.

[2]

MA-301/2311

2. Prove that an analytic function with constant modulus is constant.

3. Evaluate :

$$\int_C \frac{z^2 e^z}{(z-1)^3} dz$$

where C is the circle $|z| = 2$.

4. Obtain Taylor's series to represent the function

$$\frac{z+5}{(z+1)(z+2)} \text{ in the region } |z| < 1.$$

5. Determine the order of convergence of Regula-Falsi method.

6. Using Newton-Raphson method, find a real root of $e^x \cos x = 1.2$ correct to three decimal places.

7. Solve the following equations by Gauss elimination method :

$$3x + y + 2z = 3$$

$$2x - 3y - z = -3$$

$$x + 2y + z = 4$$

[3]

MA-301/2311

8. Prove that :

$$\nabla = \Delta E^{-1} = E^{-1} \Delta = I - E^{-1}$$

9. Calculate $\int_0^{\pi/2} e^{\sin x} dx$ correct to four decimal

places by Simpson's $\frac{1}{3}$ rule.

10. Solve the following equation by Euler's method :

$$\frac{dy}{dx} = x^2(1+y)$$

$$y(1) = 1 \text{ at } x = 1.1 \text{ with } h = 0.01.$$

11. A dice is thrown twice. What is the probability that at least one of the two throws come up with the number 3 ?

12. The first three moments of a frequency distribution about $x = 2$ are 1, 16 and -40 respectively. Determine mean and variance.

P. T. O.

13. For the data as given below :

x	y
1	14
2	27
3	40
4	55
5	68

fit the curve $y = a + bx + cx^2$.

14. Six coins are tossed 6400 times. Using the Poisson distribution, what is the approximate probability of getting six heads two times ?
15. A sample of 900 values has the mean 3.4 cm. Is the sample drawn from a large population of mean 3.25 cm and standard deviation 2.61 cm ? (Use 5% level of significance).

Section—B

(Long Answer Type Questions)

Note : Attempt any two questions. Each question carries 30 marks. $2 \times 30 = 60$

1. If the probability that an individual suffers a bad reaction from a certain injection is 0.001,

determine the probability that out of 2000 individuals :

- (i) exactly 3
- (ii) more than 2 individuals
- (iii) none
- (iv) more than one individual will suffer a bad reaction.

2. Find the coefficient of regression and the equation of lines of regression for the following data :

Price (x)	Demand (y)
78	125
89	137
97	156
69	112
59	107
79	136
68	123
61	108

P. T. O.

3. If $f(z) = u + iv$ is an analytic function of $z = x + iy$ and :

$$u - v = \frac{(e^y - \cos x + \sin x)}{(\cosh y - \cos x)}$$

find $f(z)$ subject to the condition

$$f\left(\frac{\pi}{2}\right) = \frac{3-i}{2}.$$

4. Apply calculus of residues to prove that :

$$\int_{-\infty}^{\infty} \frac{\sin x \, dx}{(x^3 + 4x + 5)} = -\frac{\pi}{e} \sin 2.$$

5. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x=1$ from the following table :

x	y
1	198669
2	295520
3	389418
4	479425
5	564642
6	644217

6. The following are the marks obtained by 492 candidates in a certain examination :

Marks not more than	Number of candidates
40	210
45	253
50	307
55	381
60	413
65	492

Find out the number of candidates :

- (i) who secured more than 48 but not more than 50 marks ?
 (ii) less than 48 but not less than 45 marks.

Do any three (3x3) (B)

1) Evaluate $\int_C \frac{z}{z^2+1} dz$ $|z+i|=1$

2) If $u+v = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2 \cos 2x}$
 find $f(z)$

3) If $f(z)$ is an analytic function of z
 prove that $(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}) |Rf(z)|^2 = 2 |f'(z)|^2$

4) Determine pole and residue at each pt
 $\frac{z^2}{(z-1)^2(z+2)}$

Do any One :- (1x6)

1) State and prove Liouville's Th

2) State & prove Cauchy Integral formula

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EC-2313

B. Tech. (EC) (Third Semester)
EXAMINATION, 2025-26
ELECTRONIC MEASUREMENTS AND
INSTRUMENTATION

Time : Two Hours

Maximum Marks : 75

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *five* questions in maximum **30** words. Each question carries 7 marks. $5 \times 7 = 35$

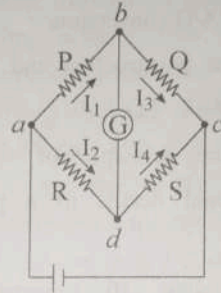
1. Define the terms :

(a) Repeatability

P. T. O.

- (b) Accuracy
 - (c) Precision
 - (d) Resolution
2. List the advantages of electronic instruments over electrical and mechanical instruments.
 3. Explain the types of Errors.
 4. Define the following :
 - (a) Transducer
 - (b) Inverse Transducer
 5. Describe the method for measurement of temperature with the use of :
 - (a) R.T.Ds.
 - (b) Thermistors
 6. Explain the different principles of working of capacitive transducers.
 7. In the Wheatstone bridge in Fig. the values of resistances of various arms are $P = 1000 \Omega$, $Q = 100 \Omega$, $R = 2,005 \Omega$ and $S = 200 \Omega$. The battery has an emf of 5V and negligible

internal resistance. The galvanometer has a current sensitivity of $10 \text{ mm}/\mu\text{A}$ and an internal resistance of 100Ω . Calculate the deflection of galvanometer and the sensitivity of the bridge in terms of deflection per unit change in resistance.



8. Derive the general equations for balance for A.C. bridge.
9. Explain the advantages of Hay's bridge over Maxwell's bridge.
10. Explain the advantages of electronic voltmeters over conventional type voltmeters.

P. T. O.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions in maximum **150** words. Each question carries 10 marks. $2 \times 10 = 20$

1. Explain the successive approximation method of A/D conversion.
2. Explain the construction and principle of working of a L.V.D.T.
3. Explain the principle of working and constructional details of a photo-voltaic cell.
4. The following 10 observations were recorded when measuring voltage :
41.7, 42.0, 41.8, 42.0, 42.1, 41.9, 42.0, 41.9, 42.5 and 41.8 volt.
Find : (i) mean, (ii) the standard deviation, (iii) the probable error of one reading, and (iv) the probable error of mean.

Section—C

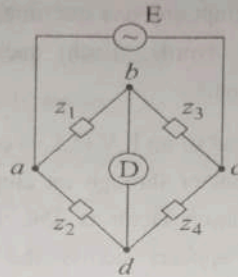
(Long Answer Type Questions)

Note : Attempt any *one* question in maximum
300 words. Each question carries
20 marks. $1 \times 20 = 20$

1. The output of an L.V.D.T. is connected to a 5V voltmeter through an amplifier whose amplification factor is 250. An output of 2 mV appears across the terminal of L.V.D.T. when the core moves through a distance of 0.5 mm. Calculate the sensitivity of L.V.D.T. and that of the whole setup. The milli-voltmeter scale has 100 divisions. The scale can be read $\frac{1}{5}$ of a division. Calculate the resolution of the instrument in mm.
2. An A.C. bridge circuit working at 1000 Hz shown in Fig. Arm *ab* : is $0.2 \mu\text{F}$ pure capacitive, Arm *bc* : is a 500Ω pure resistive, Arm *cd* : contains an unknown impedance, Arm *da* : has a 300Ω resistance in parallel with $0.1 \mu\text{F}$ capacitor.

P. T. O.

Find the R and C or L constants of cd considering it as a series circuit.



3. Derive the equations of balance for an Anderson's bridge. Draw the phasor diagram for conditions under balance. Discuss the advantages and disadvantages of the bridge.
4. Describe in detail the construction and working of an analog type storage oscilloscope. Explain the principle of secondary emission.

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PART-1 Attempt all four.

(4X1=4)

- ✓ Thermocouples are
A) Passive transducer C) both active and passive transducer
✓ B) Active transducer D) output transducer
✓ The least count of the main scale of a screw gauge is 1mm. The minimum number of divisions on its circular scale required to measure 5 μ m diameter of a wire is
✓ a) 50 (b) 200 (c) 100 (d) 500
✓ In measurement systems, which of the following static characteristics are desirable:
A) Accuracy B) Sensitivity C) Reproducibility ✓ D) All of the above
✓ A reading is recorded as 23.90 $^{\circ}$ C. The reading has
A) Three significant figures C) Four significant figure
B) Five significant figure ✓ D) None of the above

PART-2 Attempt any three questions

(3X2=6)

- ✓ The following 10 observations were recorded when measuring voltage : 41.7, 42.0, 41.8, 42.8, 42.1, 41.8, 42.6, 41.9, 42.5 and 42.8 volt. Find ✓ a) The mean, b) the standard deviation, ✓ c) the probable error of one reading, d) the probable error of mean, ✓ e) range.
✓ What are the limitations of Wheatstone bridge?
✓ Compare dc and ac bridge.
✓ If the bridge arms are connected with $R_1=10\Omega$, $R_2=5.9K$, $R_3=10K$. Find R_4 .

PART-3 Attempt any one

(1X5=5)

- ✓ Explain the Kelvin Double Bridge Method of Measurement of Low Resistances.
2) A Kelvin double bridge each of the ratio arms $P=Q=p=q=1000\Omega$. The emf of battery is 100V and a resistance of 5Ω is included in the battery circuit. The Galvanometer has a resistance of 500Ω and the resistance of the link connecting the unknown resistance to the standard resistance may be neglected. The bridge is balanced when the standard resistance $S=0.001\Omega$. determine 1) unknown resistance 2) current through the unknown resistance at balance 3) the deflection of galvanometer when the R is changed by 0.1 percent from its value at balance. Galvanometer sensitivity of 200mm/ μ A.

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EC-2322

B. Tech. (EC) (Third Semester)

EXAMINATION, 2025-26

DIGITAL ELECTRONICS

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

1. Convert and find the value of X :

(a) $(786.983)_{10} = (X)_{16}$

P. T. O.

(b) $(734)_8 = (X)_2$

(c) $(77.43)_{10} = (X)_6$

(d) $(1101.011)_2 = (X)_{10}$

2. Simplified using k -map.

$$F(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5)$$

3. Simplify and minimize using Quine Mc-Cluskey tabulation method :

$$F(A, B, C, D) = \sum m(1, 2, 3, 4, 6, 8, 9, 10, 11)$$

4. Minimize the Boolean expression :

(i) $Y = \overline{A}B + A\overline{B} + AB$

(ii) $Y = \overline{A}BC + \overline{A}B\overline{C} + A\overline{B}C + A\overline{B}\overline{C}$

5. State De-Morgan's theorem and simplify the following :

(i) $Y = (\overline{A+B+C})D$

(ii) $Y = \overline{(A+B)}\overline{C}$

6. Implement the following function :
 $F(A, B, C, D) = \sum m(0, 1, 2, 3, 4, 7, 8, 9, 11, 14, 15)$ using :
(a) 8 : 1 mux
(b) 4 : 1 mux
7. Design parallel in serial out shift register.
8. Define BCD codes with suitable example.
9. What do you mean by race around condition in JK flip-flop ?
10. Write down difference between combination and sequential digital logic circuit.

Section—B

(Short Answer Type Questions)

Note : Attempt any two questions in maximum 150 words. Each question carries 12 marks. $2 \times 12 = 24$

1. Write the difference between flip-flop and latch. Convert J.K. flip-flop into D-flip-flop.

P. T. O.

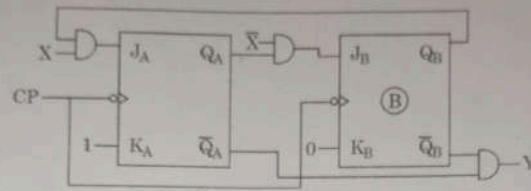
2. Design a 3-bit up/down ripple counter.
3. Draw and explain dual slope analog to digital converter.
4. Explain working principle of 3-i/p TTL NAND Gate.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question in maximum
300 words. Each question carries
20 marks. 1×20=20

1. Derive the state table and state diagram for the sequential circuit shown below :



2. Write down the difference between synchronous and asynchronous logic circuit. Explain the state diagram and state equation.
3. Draw and explain a NAND gate in Totem Pole TTL configuration.
4. Explain in detail sequential circuits using ASM charts. Also define application of sequential circuits.

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$$\overline{A}B + AB$$

P. T. O.

Department of Electronics & Communication Engg.
BIT, Bhubaneswar-751024
ODD SEM: Sessional-I (2024-25)

Class (Yr & Branch): 2 nd (EC)	Semester: 3 rd
Subject: Digital Electronics	Paper Code: 2322
Time: 60 Minutes	M.M.: 15
Student's Name:	Roll No.:

Note: All Parts are compulsory

PART-1 Attempt any 5 5x1=5

1. Convert the following:
 a. $(786.983)_{10} = (x)_{16}$ b. $(77.43)_{10} = (x)_4 = (28.64)$

2. Add $6B3 + A9C$ c. Find 2's Complement of 1011

3. Subtract using 10's complement: $76532 - 4250$ d. 1010

4. Convert the following binary number into gray code.
 a. 110111 b. 11001

5. Simplify the following Boolean expression: $AB + A(B+C) + B(B+C)$

PART-2 Attempt any 2 2x3=6

6. Simplified using K-Map: $F(A,B,C,D) = \sum m(1,3,7,11,15) + d(0,2,5)$

7. Implement and explain 1 bit magnitude comparator.

8. Implement and explain 4 bit adder/subtractor circuit.

9. Implement 8:1 MUX using 4:1 MUX.

PART-3 Attempt any 1 1x4=4

10. Simplify the following function using tabulation method
 $F(A,B,C,D) = \sum m(0,1,2,3,4,6,8,9,10,11)$

11. Implement the following function $F(A,B,C,D) = \sum m(0,1,2,3,4,7,8,9,11,14,15)$ Using:
 a. 8:1 MUX b. 4:1 MUX

12. Implement and explain Full adder circuit in details.

*****ALL THE BEST*****

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EC-2321

B. Tech. (Third Semester)
EXAMINATION, 2025-26
SEMICONDUCTOR MATERIAL AND DEVICES
Time : Two Hours
Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A
(Very Short Answer Type Questions)

Note : Attempt any seven questions. Each question carries 8 marks. 7×8=56

1. How many atoms are assigned to the unit cell of a (i) primitive, (ii) face centre, (iii) body centre cubic lattice ?

P. T. O.

2. Draw the plane for the following Miller indices :

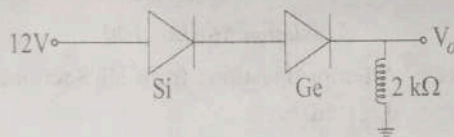
(i) (2, 1, 2)

(ii) (1, 2, 4)

(iii) (0, 1, 2)

(iv) (0, 1, 1)

3. Determine the current flowing in the circuit shown in figure :



Also calculate output voltage V_o .

4. (a) What is the momentum of photon if the de-Broglie wavelength is 5.4 m^2 ?

(b) Prove that :

$$h\nu = h\nu_0 + \frac{1}{2}m_e v_{\text{max}}^2$$

5. What do you understand by Photoelectric Effect ? State the laws of photoelectric effect.
6. Explain the working of bridge rectifier circuit.
7. With the help of suitable diagram explain varactor diode.
8. With the help of suitable diagram explain characteristic curve of DIAC.
9. In short define IGBT.
10. Briefly explain the solar cell.

Section—B

(Short Answer Type Questions)

Note : Attempt any two questions. Maximum word limit is 150. Each question carries 12 marks.

$2 \times 12 = 24$

1. What do you understand by imperfection in solid ? Explain.
2. With the help of suitable diagram explain the working of triac.

P. T. O.

3. What do you understand by LDR ? Explain its construction and working.
4. What do you understand by luminescence ? Explain in detail.

Section—C

(Long Answer Type Questions)

Note : Attempt any one question. Each question carries 20 marks. $1 \times 20 = 20$

1. What do you understand by A.P.F. ?
Calculate APF for :
 - (i) Simple cubic structure
 - (ii) BCC structure
 - (iii) FCC structure
2. Explain with diagrams the construction and working of SCR. Also give its merits and demerits.

3. Prove that built in potential V_{bi} in P-N junction is given by :

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_B}{n_i^2} \right)$$

Also define relationship between width and built in potential.

4. Write short notes on the following :

- (i) Optical absorption
- (ii) Optical transistor
- (iii) Hall effect
- (iv) de-Broglie hypothesis

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Roll No. [Total No. of Pages : 4]

EC-2321

B. Tech. (Third Semester)
EXAMINATION, 2024-25
SEMICONDUCTOR MATERIAL AND
DEVICES

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions in about
30 words. Each question carries
8 marks. $7 \times 8 = 56$

1. What do you understand by a solid ?
Explain with a suitable diagram.

P. T. O.

[2]

EC-2321

2. What is Frenkel defect ? Explain.
3. With the help of suitable diagram, explain solar cell.
4. What is photoconductivity ? Explain.
5. Explain transition and diffusion capacitance.
6. What do you understand by Miller indices ? Explain.
7. Explain centre tap full wave-rectifier.
8. Explain the working of JFET.
9. With the help of a suitable diagram, explain TRIAC.
10. What is luminescence ? Explain.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions in about
150 words. Each question carries
12 marks. $2 \times 12 = 24$

1. With the help of a suitable diagram, explain the construction and working of MESFET.

2. Prove that built-in potential in p - n junction is given by :

$$V_{bi} = \frac{kT}{q} \ln \frac{(N_A N_D)}{n_i^2}$$

3. With the help of a suitable diagram, explain Schottky diode.
4. With the help of a suitable diagram, explain IGBT.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question in about 300 words. Each question carries 20 marks. $1 \times 20 = 20$

- (1.) With the help of a diagram, explain SCR in detail.
- (2.) Explain the construction and working of LED. Also give its application, merits and demerits.

P. T. O.

- (3.) What do you understand by charge carriers in a semiconductor ? Determine drift and diffusion current.
- (4.) Write short notes on the following : *
- (i) Bonding force and energy band in solid
 - (ii) Carrier life time
 - (iii) Optical fiber
 - (iv) Heterojunction

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EC-301/2321

B. Tech. (EC) (Third Semester)

EXAMINATION, 2023

**SEMICONDUCTOR MATERIAL AND
DEVICES**

Time : Three Hours

Maximum Marks : 100

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any *ten* questions. Each question carries 4 marks. $10 \times 4 = 40$

1. What is lattice and unit cell ? Explain with diagram.

P. T. O.

[2]

EC-301/2321

2. Define Miller indices. Referring to the unit cell, draw the following Miller planes : (100), (231)
- ✓ 3. Define diffusion of carriers.
- ✓ 4. Describe atomic packing factor.
- ✓ 5. What is Hetero-Junction ? Explain.
6. Explain the avalanche photodiode.
- ✓ 7. How many atoms are assigned to the unit cell of a centre cubic lattice ?
8. Give the properties of Miller indices.
- ✓ 9. What is the momentum of photon if the de Broglie wavelength is 5.4 m ?
- ✓ 10. What is electroluminescence ?
- ✓ 11. What are the two mechanisms of breakdown in P-N Junction ? *ZEY. / A.S.*
- ✓ 12. Explain the process of photoelectric emission.
- ✓ 13. With the neat diagram explain solar cell.
- ✓ 14. What is Hall effect ? Explain. Also give its application.
- ✓ 15. Draw the V-I characteristics of MOSFET and explain it.

$\sim \times \frac{4}{3} \pi r^3$

$\frac{M_o}{V}$

$\lambda = \frac{h}{mv}$
kg m/s

Section—B

(Long Answer Type Questions)

Note : Attempt any *three* questions. Each question carries 20 marks. $3 \times 20 = 60$

1. Define the following :

- (a) Compound semiconductor
- (b) Miller indices
- (c) Bragg's law
- (d) Imperfection in crystals

2. (a) What is photoelectric emission ? How is electron emission effect if (i) frequency (ii) the intensity of the incident radiation are increased ? Name the material which is suitable for photo emission.

3. (b) Explain photoconductivity in a semiconductor material.

3. (a) What do you understand by Zener diode ? Explain its following applications :

- (i) Meter protection
- (ii) Switching operation

P. T. O.

(b) With the help of a suitable diagram explain the construction and working of varactor diodes.

4. What do you understand by SCR ? Explain. Give its construction. Also define its volt ampere characteristics and dynamic characteristics.

5. What is LED ? Give its principle of working, construction, circuit symbols, merits, demerits and application.

6. Write short notes on the following :

- (a) LDR
- (b) Photovoltaic or Solar cell
- (c) IGBT
- (d) Carrier time